DSA ki Baithak

CODE BAITHAK

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1. Fibonacci Number

$$\left[f(n) = f(n-1) + f(n-2) \right] \quad n > 1$$

0-based $\left[\begin{array}{l} f(0) = 0 \\ f(1) = 1 \end{array} \right]$

$f(3) = f(2) + f(1)$

$f(4) = f(3) + f(2)$

$f(2) = f(1) + f(0) = 1 + 0 = 1$

Input:

$n = 4$

0, 1, 1, 2, 3, 5, 8, ...

0 1 2 3 4

Output:

3

1

2

$n = 0$, $n = 1$, $n > 1$

a → b →

~~a~~ ~~b~~ ~~sum~~ ~~sum~~ ~~sum~~

0, 1, 1, 2, 3, 5, ...

n = 4

0, 1, 1, 2, 3, 5, - - -

result

a b

a b

result

one = ~~0~~ + ~~1~~ 2

two = ~~1~~ + ~~2~~ 3

result = ~~0~~ + ~~2~~ 3

i = ~~2~~ ~~3~~ ~~4~~ 5

```
if (n <= 1)
    return n

one = 0
two = 1
result

for (int i=2; i<=n; i++) {
    result = one + two
    one = two
    two = result
}
```

two

2. N-th Tribonacci Number

$$\left[T(n+3) = T(n) + T(n+1) + T(n+2) \right]$$

Input :

$n = 4$

Output :

4

3. Inverse of Number

① ✓

② digit → place

place → digit

3	2	1	4	5
5	4	3	2	1

1	2	5	4	3

→

$$\begin{array}{r} 10000 \\ 2000 \\ 500 \\ 40 \\ + \quad 3 \\ \hline \end{array}$$

Input :

n = 32145

Output :

12543

$$1\times10^4 + 2\times10^3 + 3\times10^0 + 4\times10^1 + 5\times10^2$$

5

n = ~~32145~~ ~~3214~~ ~~321~~ ~~32~~ ~~3~~ 0

place = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6

ans = $1 \times 10^4 + 2 \times 10^3 + 3 \times 10^0 + 4 \times 10^1 + 5 \times 10^2$

digit = ~~5~~ ~~4~~ ~~1~~ ~~2~~ 3

ans = 10000
2000
3
40
500

12543

```
ans = 0
place = 1
while(n>0) {
    digit = n%10
    ans += place * pow(10, digit)
    place++
    n/=10
}
```

4. GCD/ HCF of Two Numbers

① ✓ ② ✓

$a = 5$, $b = 10$ → 5

5%

min(20, 28) → 20

$b = 28 \rightarrow$ 1, 2, 4, 7, 14, 28

$a = 20 \rightarrow$ 1, 2, 4, 5, 10, 20

Input :

$a = 20$, $b = 28$

Common factors = 1, 2, 4 → GCD / HCF

Output :

4

x → $b \% x == 0$ && $a \% x == 0$

1, 2, 3, 4 — (4)

1, 2, 3, 4 → (1)

Long Division Method of finding GCD :

(Euclidean Algorithm)

gcd (20, 28) =

20) 28 (
− 20
8) 20 (2
− 16
4) 8 (2
− 8
x

3

gcd

divisor

28) 20 (0
− 0
20) 28 (

```
while (divid % divl ≠ 0) {
    rem = divid % divsor
    dividend = divsor
    divsor = rem
}
```

5. LCM of Two Numbers

$12 \% 18 == 0$

start → max(a, b)

end → a*b

a = 12 → 12, 24, 36, 48, . . .

b = 18 → 18, 36, 54, ...

Input:

a = 12, b = 18

x

1, 2, 3, 4

36 → LCM

a*b

Output:

36

1 →

if (x % a == 0 && x % b == 0)

return x

for(

GCD - LCM Formula :

$$\left[a * b = \gcd(a,b) * \operatorname{lcm}(a,b) \right]$$

$$\therefore \quad \operatorname{lcm}(a,b) = \frac{a * b}{\gcd(a,b)}$$